Chapter 1

Background

Approximate Bayesian Computation methods are a family of simulation algorithms designed to produce samples from an approximate posterior distribution. Such methods were proposed as an alternative to standard Monte Carlo techniques to make statistical inference possible for a class of applied problems where the likelihood function cannot be evaluated. This chapter summarizes some of the most influential ABC approaches, particularly those directly related to the content introduced in the following chapters. Section 1.1 considers a simple introductory example of estimation via ABC. Then, in Section 1.2, we explore more deeply the nature of each of the input elements that define an ABC sampler (such as the summary statistics), taking the opportunity to provide some guidelines in regards to the question of how to specify them in practice. To conclude, several ABC approaches are reviewed in Section 1.3.

1.1 Introductory example

A rejection-ABC algorithm was previously described in the Introduction section. We now turn our attention to a slightly more sophisticated version of ABC, known as ABC importance sampling. The underlying principle is common to both implementations – that is, to avoid direct evaluation of the likelihood by matching synthetic samples to an observed dataset. But rather than simply rejecting or accepting a candidate parameter, the latter algorithm generalizes the former by assigning an importance weight to each approximate posterior sample.

The importance weights are defined by three interrelated components. First, a function to compute *summary statistics* from data, $S(\boldsymbol{X})$, is chosen to reduce the dimension of the objects being matched. Second, a *distance metric*, $D(\boldsymbol{s}, \boldsymbol{s}_{\text{obs}}) = \|\boldsymbol{s} - \boldsymbol{s}_{\text{obs}}\|$, is employed to measure the degree of similarity between a synthetic sample $\boldsymbol{s}$ and the observed summary statistics $\boldsymbol{s}_{\text{obs}}$. Third, a *weighting function*, $K_h(d)$, which is controlled by a bandwidth parameter h, sets the importance weights according to $D(\boldsymbol{s}, \boldsymbol{s}_{\text{obs}})$ – the bigger the distance the smaller the weight.

Algorithm 1 produces weighted samples from the ABC posterior approximation given by

$$\pi_{ABC}(\boldsymbol{\theta}|\boldsymbol{s}_{\text{obs}}) = \int K_h(\|S(\boldsymbol{x}) - \boldsymbol{s}_{\text{obs}}\|)p(\boldsymbol{x}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})d\boldsymbol{x}. \tag{1.1}$$

It follows from Eq. (1.1) that if the analyst assigns weight proportional to one when the observed summary statistics are perfectly reproduced ($\|S(\boldsymbol{x}) - \boldsymbol{s}_{\text{obs}}\| = 0$) and zero otherwise, then Algorithm 1 generates exact samples from the partial posterior distribution $\pi(\boldsymbol{\theta}|\boldsymbol{s}_{\text{obs}})$. However, this choice makes the sampler hopelessly inefficient, as the probability of generating an acceptable match becomes prohibitively small. In general, specifying those three components is a delicate, non-trivial task.

Algorithm 1 ABC Importance Sampling (vanilla version)

Inputs:

- A target posterior density $\pi(\boldsymbol{\theta}|\boldsymbol{X}_{\text{obs}}) \propto p(\boldsymbol{X}_{\text{obs}}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})$, consisting of a prior distribution $\pi(\boldsymbol{\theta})$ and a procedure for generating data under the model $p(\boldsymbol{X}_{\text{obs}}|\boldsymbol{\theta})$.
- An integer $\tilde{N} > 0$.
- An observed vector of summary statistics $\boldsymbol{s}_{\text{obs}} = S(\boldsymbol{X}_{\text{obs}})$.
- A kernel function $K_h(u)$ and scale parameter $h > 0$.

Sampling:
For $i = 1, \ldots, \tilde{N}$:

1. Generate $\boldsymbol{\theta}^{(i)} \sim \pi(\boldsymbol{\theta})$ from the prior.
2. Generate $\boldsymbol{X}^{(i)} \sim p(\boldsymbol{X}|\boldsymbol{\theta}^{(i)})$ from the likelihood.
3. Compute the summary statistics $\boldsymbol{s}^{(i)} = S(\boldsymbol{X}^{(i)})$.
4. Assign $\boldsymbol{\theta}^{(i)}$ the weight $w^{(i)} \propto K_h(\|\boldsymbol{s}^{(i)} - \boldsymbol{s}_{\text{obs}}\|)$.

Output: A set of weighted parameter vectors $\{(\boldsymbol{\theta}^{(i)}, w^{(i)})\}_{i=1}^{\tilde{N}} \sim \pi_{ABC}(\boldsymbol{\theta}|\boldsymbol{s}_{\text{obs}})$.

Hereafter we explore – in a graphical, intuitive level – a simulated toy example to give the reader a gentle introduction to the ABC machinery, with main focus on its operational aspects. For the sake of illustration, we consider a simple model characterized by a sequence of *dependent* stochastic events. Precisely, each observation, x_i, $i = 1, \ldots, N$, is obtained by (a) drawing a sample from a normal distribution with mean θ and variance 1; and (b) if the value obtained in (a) is negative, multiplying it by -1 with probability $p = 0.5$. The fundamental feature being that the action, or lack of thereof, in step (b) is conditional on what happened in (a). Alternatively, this model can described by the following equations:

$$x_i = \begin{cases} |y_i|, & \text{with probability } p = 0.5. \\ y_i, & \text{with probability } 1 - p = 0.5. \end{cases}$$

$$y_i \sim N(\theta, 1).$$

We assigned a standard Gaussian prior for θ, which completes the (prior predictive) data generative process under consideration. Conditioned on $\theta = -1$, which we take as the "true" parameter value, we simulated 50 samples from the model to act as the "observed" dataset. The density function $p(X|\theta = -1)$ is depicted in the bottom-right plot in Figure 1.1.

The likelihood can be easily evaluated here. However, in more elaborated constructions, where the number of probabilistic events are substantially higher, the conditional structure becomes overwhelmingly complex, rendering the likelihood intractable (Luciani et al., 2009; Tanaka et al., 2006). The epidemiological model analyzed in Chapter 2, for example, assumes that, in each time step, several random events can take place for each member of the population, depending on their current status – subjects can get infected, recover, die, etc. The number of distinct ways in which the population could have reached its final, observed state is extraordinarily large.

In this study, the univariate data is summarized by its sample mean, $\bar{x}$, and standard deviation, $\hat{\sigma}$. The Euclidean distance and the Epanechnikov kernel were chosen as the distance metric and the weighting function respectively. Algorithm 1 was then used to generate $\tilde{N} = 1000$ approximate samples from the ABC posterior distribution for a number of tolerances h.

The upper-left plot in Figure 1.1 show the objects generated by Steps 1 to 3 of Algorithm 1. Each parameter $\theta^{(1)}, \ldots, \theta^{(\tilde{N})}$ is associated to two points in the plot – one representing the mean and the other the standard deviation of the corresponding synthetic sample. The relationship between the parameter of interest and the summary statistics forms an interesting pattern. When θ is large, the probability of generating a negative sample y becomes negligible, so the density function effectively becomes a single normal distribution centered at θ. In this region, the sample mean is sufficient for θ, while the standard deviation becomes uninformative – notice the cluster formed around the true standard deviation $\sigma = 1$. On the other side of the coordinate axis, when θ is, say, less then -3, then each generated data point is reflected with probability 0.5, turning the density function into a mixture of two normals with opposite means. In this case, $\bar{x}$ estimates $E(X|\theta) \approx 0$, carrying no useful information about θ. The standard deviation, however, assumes a linear, negative relationship with θ.

The vertical lines represent the "observed" summary statistics. We see that the model can only reproduce $\hat{\sigma} \approx 1.5$ if θ is in a neighborhood of -1. The plots in the right-hand-side explore the relationship between θ and the computed Euclidean distances $\|\boldsymbol{s}^{(i)} - \boldsymbol{s}_{\text{obs}}\|$. Again, the observed patterns suggests that the distance can only be small if $\theta \approx -1$.

Step 4 of Algorithm 1 is illustrated by the following plots of Figure 1.1. We first set h so that 25% of the candidate parameters are assigned positive weights. As the majority of the original samples were discarded, the clouds of points now cluster more closely around the vertical lines. As h is further reduced, the samples get even closer the space over which the exact posterior is supported. On the other hand, the number of accepted samples also shrinks, which prompts an increase in the Monte Carlo error. This trade-off, which is a well-known feature of ABC methods, is observed in the bottom-left plot of Figure 1.1.

1.2 Constructing an ABC sampler in practice

For any given model, there are numerous ways to setup an ABC sampler. For a more extensive introduction to the ABC literature, see Sisson et al. (2017a); Beaumont (2010); Csillery et al. (2010); Lopes and Beaumont (2010); Bertorelle et al. (2010); Sisson and Fan (2011); Turner and Van Zandt (2012); Blum et al. (2013). As illustrated in our toy example, defining an appropriate weighting system is paramount to alleviate the ABC approximation error.

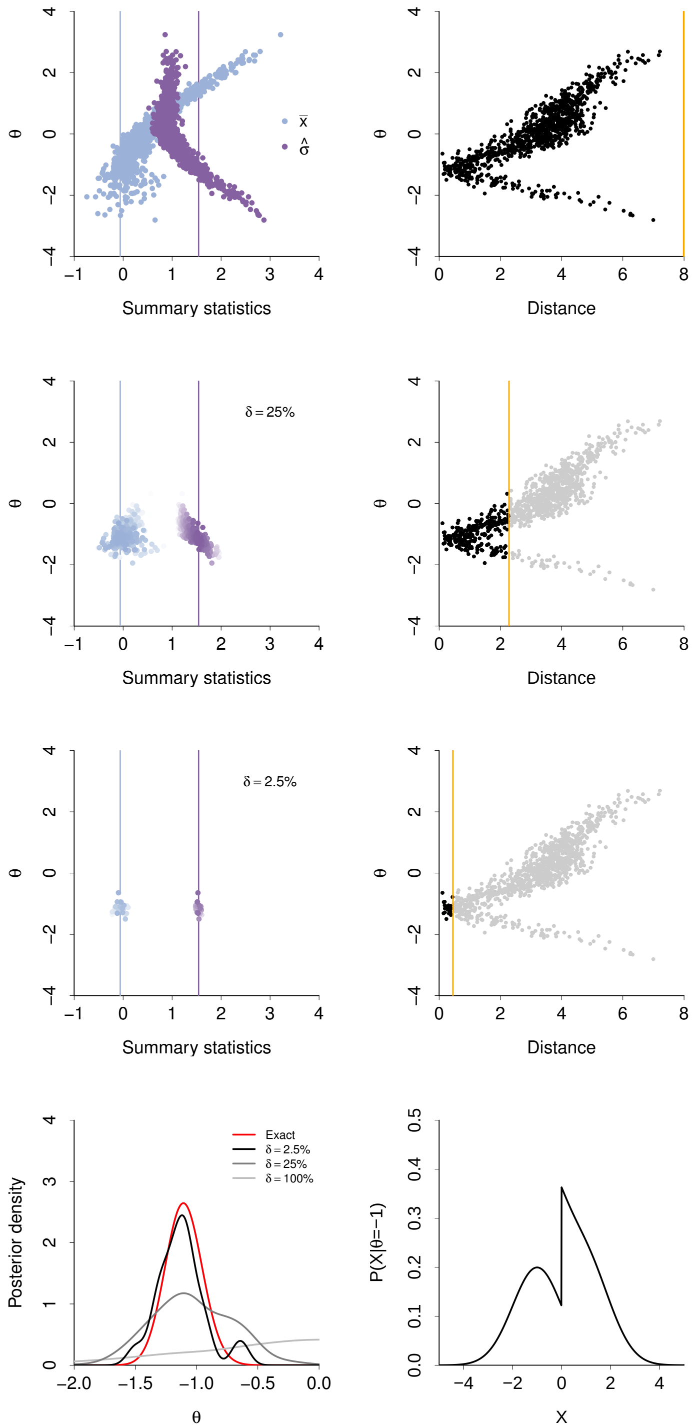

Figure 1.1: The upper-left plots show the synthetic (points) and the observed (vertical lines) summary statistics for different acceptance ratios δ. The color intensity reflects the importance weights (calculated from the Epanechnikov kernel function), with plain white representing zero weight. The corresponding plots in the right-hand-side present the Euclidean distances between synthetic and observed summary statistics. Samples in light gray were assigned weight zero (and therefore rejected). The orange line represents the tolerance h induced by δ. The bottom-left plot compares the exact (red) and the ABC approximate posteriors (in gray). The bottom-right plot shows the density function for data, conditioned on the "true" parameter value $\theta = -1$.